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13. ENVIRONMENTAL MONITORING AND AUDIT REQUIREMENTS

13.1 Introduction

13.1.1.1 This section elaborates the requirements of environmental monitoring and audit (EM&A) for the construction and operational phases of the Project, based on the assessment results of the various environmental issues. Details of the requirements are provided in a stand-alone EM&A Manual.

13.1.1.2 The purpose of the EM&A programme is to ascertain and verify the assumptions implicit to, and accuracy of, EIA study predictions. The EM&A programme includes the scope of the EM&A requirements for the Project to ensure compliance with the EIA study recommendations, to assess the effectiveness of the recommended mitigation measures and to identify any further need for additional mitigation measures or remedial action. The following sections summarise the recommended EM&A requirements for this Project. The Implementation Schedule of Recommended Mitigation Measures is presented in the stand-alone EM&A Manual.

13.2 Air Quality

13.2.1.1 EM&A for potential air quality impacts should be conducted during construction phase so as to check compliance with the legislative requirements. Details of the monitoring and audit programme are contained in a stand-alone EM&A Manual.

13.2.1.2 Regular site audit for potential air quality impact, continuous dust monitoring is recommended to be conducted during the entire construction phase of the Project so as to ensure the air quality mitigation measures, including those stipulated in *Air Pollution Control (Construction Dust) Regulation* are implemented in order and to ensure that no nearby air sensitive receivers (ASRs) will be subject to adverse air quality impact.

13.2.1.3 No adverse air quality impact is anticipated during the operation of the Project. No EM&A would be required during the operation phase of the Project.

13.3 Noise

13.3.1.1 Noise monitoring is recommended as part of the EM&A programme for the construction phase of the Project to check compliance with the construction noise criteria. The implementation of the recommended mitigation measures for construction activities as part of the EM&A programme, with reference to the latest implementation schedule in Construction Noise Management Plan (CNMP) to be submitted and agreed with Director of Environmental Protection (DEP).

13.3.1.2 There would be no adverse road traffic noise impact anticipated from the Project with implementation of the proposed mitigation measures. At-source mitigation measure (i.e. low noise road surfacing and vertical / cantilevered noise barrier) is recommended on part of the Project roads, while at-receiver mitigation measures (e.g. acoustic windows / acoustic balcony) are recommended at some of the planned residential developments. Road traffic noise levels should be monitored at the selected representative noise sensitive receivers (NSRs) located in the vicinity of the recommended at-source mitigation measures, during the first year of road opening. Details of the EM&A programme are provided in a stand-alone EM&A Manual.

13.3.1.3 With proper implementation of noise control measures, with reference to **Section 4**, no adverse fixed noise impact would be anticipated from the proposed administration building and several portal plant building / rooms. The specification, mitigation measures, shall make reference to the latest implementation schedule as stated in the Fixed Noise Sources Management Plan (FNMP) and corresponding quantitative fixed noise assessment to be submitted and agreed with DEP. Nevertheless, as part of the design process, monitoring of operational noise from the proposed fixed plants during the testing and commissioning stage and monitoring and audit requirements would be required in the recommendations in the FNMP to verify the compliance of the EIAO-TM criteria.

13.4 Water Quality

13.4.1.1 With full implementation of the mitigation measures such as good site practices, storm water pollution control plan, proper construction method, etc, no adverse impact and cumulative impact are anticipated during both the construction and operation phases of the Project. In order to ensure effectiveness of the implemented mitigation measures, regular water quality monitoring in Kam Tin River and site audit are recommended during the construction phase. Details of the monitoring and audit requirements are provided in the stand-alone EM&A Manual for the Project.

13.5 Waste Management Implications

13.5.1.1 It is the Contractor's responsibility to ensure that all wastes produced during the construction of the Project are handled, stored and disposed of in accordance with good waste management practices, relevant legislation and waste management guidelines.

13.5.1.2 Waste materials generated during construction activities, such as construction and demolition (C&D) materials, are recommended to be audited at regular intervals (e.g. monthly) to ensure that proper storage, transportation and disposal practices are being implemented. This would ensure the waste generated would be properly disposed of. The Contractor should be responsible for the implementation of any mitigation measures to minimise waste or mitigate problems arisen from waste materials.

13.5.1.3 A Waste Management Plan (WMP), as a part of the Environmental Management Plan in accordance with *ETWB TC(W) No. 19/2005* should be prepared and submitted to the Project Manager for approval. The recommended mitigation measures in this EIA Report should form the basis of the WMP. The monitoring and auditing requirement stated in *ETWB TC (W) No.19/2005* should be followed with regard to the management of C&D materials.

13.5.1.4 Adverse environmental impacts associated with handling, storage and disposal of waste are not expected from the operation of the Project with the implementation of good waste management practices. Therefore, waste monitoring and audit programme for the operation phase of the Project would not be required.

13.6 Land Contamination

13.6.1.1 Remediation works, if necessary, would be carried out based on the recommended further works outlined in **Section 7** of the EIA Report. Mitigation measures as recommended in the future approved Remediation Action Plan should be

implemented during the remediation works. The EM&A requirements should be carried out in the form of weekly site inspection to ensure the recommended mitigation measures are properly implemented.

13.7 Ecology

13.7.1.1 Ecological resources were identified as requiring mitigation measures, as outlined in **Section 8**. These include wetland and woodland compensation, pre-construction survey with associated transplantation/translocation of species of conservation importance and construction time restrictions. It is recommended that these measures be monitored and audited by qualified ecologist(s) with at least 7 years of relevant experience throughout the construction phase. This will ensure proper implementation and, where appropriate, monitor the performance of the proposed mitigation measures. Furthermore, a monthly site audit should be carried out throughout the construction phase to ensure that the recommended ecological mitigation measures, as well as pollution control measures, are properly implemented. In case of non-compliance, the contractor should be informed and instructed to strengthen the proposed measures accordingly. Detail of mitigation measures are provided in the standalone EM&A Manual.

13.7.1.2 EM&A programmes were recommended to ensure compliance of the potential air quality, noise impacts and water quality impact respectively (e.g. potential dust emission during construction phase, potential noise exceedance from construction noise, and potential water pollution). Monitoring requirements are further stated in the corresponding sections of the EM&A Manual.

13.8 Fisheries

13.8.1.1 The pond areas to be lost do not account for a significant proportion of the total fishpond area in Hong Kong, therefore the overall fisheries impact would range from negligible to low. Indirect impacts arising from construction activities and operation of the proposed development would be properly mitigated through standard practices. Upon the implementation of recommended mitigation measures, no adverse impact on fisheries resources is anticipated from the construction and operational phase of the Project. Hence, monitoring of fisheries is not recommended.

13.9 Landscape and Visual

13.9.1.1 The EIA has recommended landscape and visual mitigation measures to be undertaken during construction and operation phases of the Project. The design, implementation and maintenance of landscape and visual mitigation measures should be checked to ensure that they are fully realized and that potential conflicts between the proposed landscape measures and any other project works and operation requirements are resolved at the earliest possible date and without compromise to the intention of the mitigation.

13.9.1.2 The construction phase EM&A of the mitigation measures shall be carried out as part of the site audit programme. EM&A during operation phase of the Project shall be carried out within the 12-month establishment period of the landscape and visual mitigation measures by the corresponding implementation agency to ensure the proposed mitigation measures in the EIA and as depicted in the Landscape and Visual Mitigation Plan are fully implemented.

13.9.1.3 All mitigation measures proposed in the EIA and implemented by the Contractor should be audited by Registered Landscape Architect (RLA), as a member of the Environmental Team, on a regular basis to ensure compliance with the intended aims of the measures. The mitigation measures proposed should be embodied into the detailed engineering design and landscape design drawings and contract document. Site inspection should be undertaken monthly throughout the construction period. In particular, the extent of the agreed works areas should be regularly checked during the construction phase. The landscape auditor should audit the proposed mitigation measures in the EIA to ensure that they are fully implemented during construction and the 12-month establishment period during operation phase.

13.10 Cultural Heritage

Construction Phase

13.10.1.1 For built heritage, as no direct or indirect impact on built heritage is anticipated during the construction phase, environmental monitoring and audit is therefore not required.

13.10.1.2 For archaeology, the Ngau Tam Mei SAI is partially located within the Project Boundary but outside the proposed alignment. According to the latest design of the construction works arrangement, the works that will be carried out within the SAI would only be those temporary works (such as filling up the area for haul road use) that are non-intrusive in nature and would cause no sub-soil disturbance. Moreover, the methodology proposed for such works are reversible and the ground conditions will be reinstated upon construction of works. Therefore, no impact to the SAI is anticipated. The contractor should submit a method statement for the installation and removal of the temporary works within the Ngau Tam Mei SAI for agreement with AMO. The works must ensure that no disturbance to the soil matrix at Ngau Tam Mei SAI. Photographic records of the ground condition within the SAI should be taken before the execution of any works, after all works have been removed from the area, as well as at intervals during the construction phase and submitted to AMO for retention. Details of the EM&A requirements are provided in a stand-alone EM&A Manual.

13.10.1.3 According to the latest construction works arrangement, the works to be carried out within the Ngau Tam Mei SAI would only be those temporary works that are non-intrusive in nature and would cause no sub-soil disturbance. However, if the arrangement deviates from the current proposed situations, an archaeological survey-cum-excavation should be carried out to obtain archaeological data for assessing the impact before commencement of the works, in consultation with the AMO, and to design mitigation measure as appropriate.

13.10.1.4 Pursuant to the A&MO (Cap. 53), the project proponent is required to inform AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

Operation Phase

13.10.1.5 For built heritage, as no direct or indirect impact on built heritage is anticipated during the operation phase, environmental monitoring and audit is therefore not required.

13.10.1.6 No impact to archaeology is anticipated during the operation phase, as archaeological impact would have been mitigated before or during the construction phase (if any). Hence no mitigation measure is required.

13.11 Hazard to Life

13.11.1.1 Blasting activities regarding use, storage and transport of explosives should be supervised and audited by competent site staff to ensure strict compliance with the blasting permit conditions. Good safety practices and recommended design measures should be implemented to further manage and minimise the potential risks during construction phase of the Project. No environmental monitoring would be required.

13.11.1.2 No risk associated with the operation of the Project is expected, and thus no EM&A programme would be required.

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